# PROJECT AND TEAM INFORMATION

## Project Title

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| Grade Grid – A comprehensive lab evaluation platform |

## Student / Team Information

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# PROPOSAL DESCRIPTION

## Motivation

In our college in a class there are around 70-80 students, and 1 teacher takes atleast 2 classes. With so many students it is messy to collect all their lab assignments and there is always a chance that students assignments might misplaced. Moreover it is difficult for students to know how their assignment was evaluated, what were the criteria on which they got less marks than desirable and why the teacher gave them such marks. Also web development is an important, a subject a student of btech should know how to implement.

This project aims to solve both of these problems by creating a website where students can upload their assignments, clearly see what are the criteria's they are judge on and see the remarks of the teacher and understand why they got their current marks. It will also simplify the process of evaluation for the teacher. We will also gain practical knowledge on MERN stack by creating this project.

## State of the Art / Current solution

There are two website which are similar to what we are creating namely Autolab, Autograder.io. Autolab is an automated grading platform used by universities to evaluate programming and lab assignments. Students submit their work online, and the system runs predefined test cases to check correctness. It automatically calculates marks and provides instant feedback. Instructors can customize grading scripts and scoring rules. It supports multiple programming languages and large classes.

Autograder.io is an open-source web application for automatically grading student assignments. Students upload their submissions, and the system evaluates them using instructor-written test cases. It generates scores and feedback without manual checking. The platform is lightweight and suitable for managing assignments in programming or technical labs. It can handle many students efficiently.

## Project Goals and Milestones

The primary goal is to create a full stack website which will help teachers by giving them additional tools to be able to create criteria's, give remarks, see which class has what deadline but also help students see on what basis were their assignments judged on, and why were they given such marks.

Milestone 1:

- Simple login and signup page for both students and teachers.
- Create basic prototype where students will be able to upload their assignments.
- Teachers will be able to give marks and remarks.

Milestone 2 :

- Refine the frontend using necessary libraries.
- Add calendar, tool to create different criteria's and other useful features.
- Create database to store all of this data.
- Connect frontend to backend using Api.

Milestone 3 :

- Refine the system and look for errors.
- Give grade automatically based on marks given
- Add features to exports marks of all students to an excel or csv file.
- Host the website.

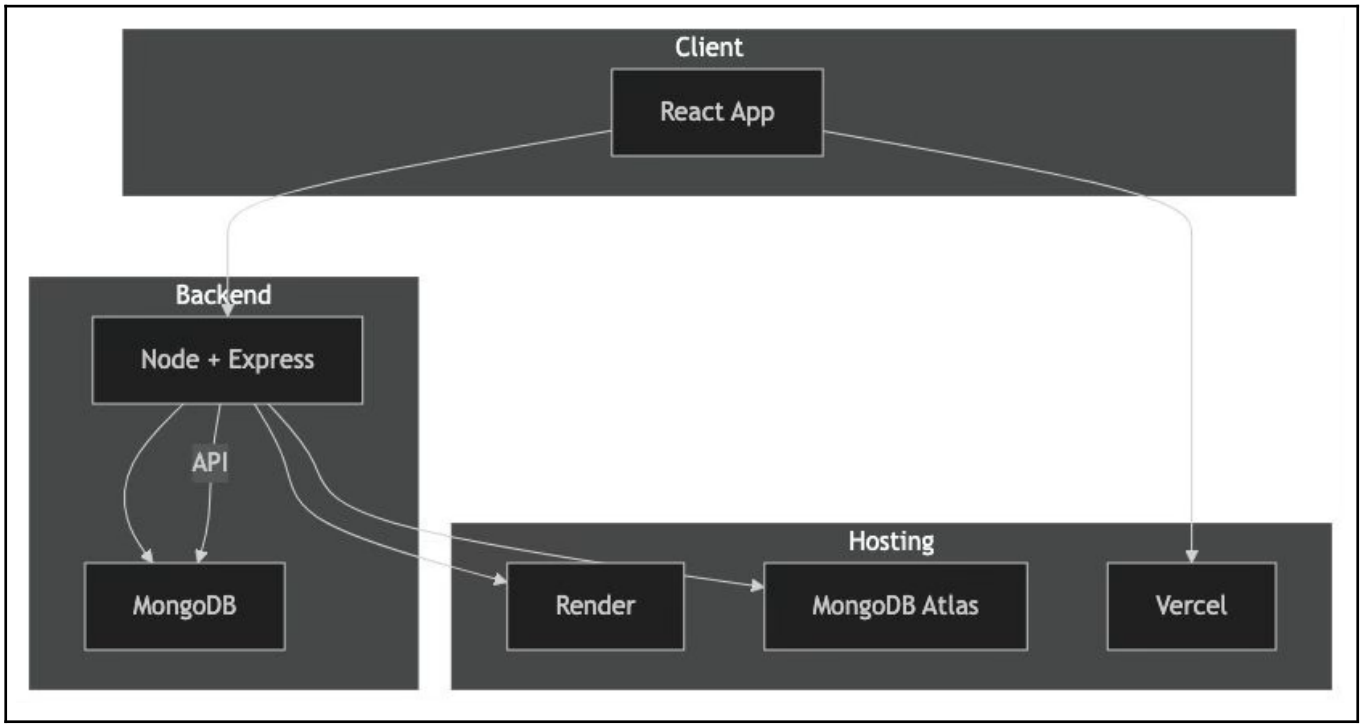
## Project Approach

We will first start with the frontend, create a prototype. Then set up a database and connect the frontend and backend while simultaneously adding necessary features to the website.

- **Technologies:** As mentioned earlier we will be using MERN stack to build this project, so MongoDB [1] a nosql database to store data. Express.js [2] to create API's and hand API request. React [3] to create interactive frontend for the users to interact and finally Node.js [4] to handle JavaScript on the sever side.
- **Tools:**
    - **Git [5] & GitHub [6]:** For version control and collaborative development.
    - Vercel [7]: To host our frontend.
    - Render [8]: To host our backend

These are the tools and technologies required for the creation of our project.

## System Architecture (High Level Diagram)




## Project Outcome / Deliverables

The primary deliverable will be a website for evaluation of lab assignments with helpful features for both students and teachers. The project will also produce:

1. The complete, well-documented source code for our project, managed in a public GitHub repository.
2. A final project report detailing the system's architecture, implementation challenges, key design decisions, and a user guide on how to build and run the OS.

## Assumptions

- At least for the initial deployment it is assumed that there will be limited number of people will be using it.
- The application will run on standard desktop/laptop devices.

## References

[1] MongoDB : https://www.mongodb.com/docs/

[2] Express.js: https://expressjs.com/

[3] React: https://react.dev/

[4] Node.js: https://nodejs.org/docs/latest/api/

[5] Git: https://git-scm.com/docs

[6] GitHub: https://docs.github.com/

[7] Vercel: https://vercel.com/docs

[8] Render : https://render.com/docs/

[9] Fritz Schneider, Thomas Powell, JavaScript – The Complete Reference, 2017, 3rd Edition, McGraw Hill.